主修---硬件與好處 (中央處理器以外)

Name: _____ ( ) Class: _____

2012

Peter buys two computers with the following specifications.

Type	Notebook Computer	Tablet Computer
CPU	2.80 GHz	1 GHz
Main memory	2 GB	512 MB
Storage device capacity	500 GB	64 GB
Display	13.3" LCD	9.7" LED
Input device	Keyboard	Virtual keyboard
WiFi	Yes	Yes
LAN port	Yes	No
USB port	Yes	Yes
Operating system	Linux (Fedora)	Android

(a) Name the storage devices commonly used in the two computers.

Notebook computer: _____

Tablet computer: _____

(2 marks)

(b) Use Part (a) and the specifications above to justify the following arguments.

(i) The tablet computer has longer battery operating time.

_____

_____

(ii) The tablet computer needs a shorter startup time.

_____

_____

(iii) The tablet computer is not suited to typing for a long period of time.

_____

_____

(3 marks)

(c) Peter wants to connect a printer to the tablet computer. Suggest **two** types of connectivity and describe the benefits of each.

_____

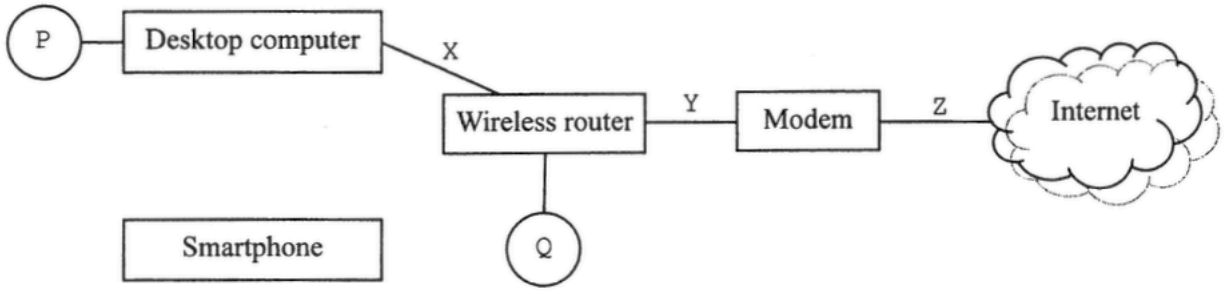
_____

_____

(2 marks)

2013

John wants to set up a local area network with a desktop computer, a smartphone and a printer at home, as shown below:



(a) (i) John connects the printer to the network at P instead of Q. What is the advantage and disadvantage of this?

Advantage: _____

_____

Disadvantage: _____

_____

(ii) The printer is equipped with RAM and ROM. When John prints a document, what will the RAM and ROM store?

(1) RAM _____

(2) ROM _____

(4 marks)

(c) Janice keeps a lot of videos and pictures of the annual Charity Runs of the past few years on her notebook computer.

(i) The notebook computer is equipped with a solid-state drive (SSD) instead of a general hard disk. Give **two** benefits to Janice of using a SSD.

_____

_____

_____

2014

In a school, Mr Li develops an interactive e-learning platform. Students can use a smart phone to access the platform and participate in some e-learning activities.

(b) Mr Li orders some desktop computers so that students who forget to bring their smart phones can still participate in the e-learning activities. Give **two** advantages of desktop computers over smart phones.

_____

_____

_____

_____

(2 marks)

(ii) Mr Li can ask students to submit the reports through either email or a school network drive. Give one advantage of each method.

Email: _____

_____

School network drive: _____

_____

(4 marks)

Don't waste the blank space, lets practice sth else :D

Ms Wong's colleague in Hong Kong, David, creates a simple web page for Ms Wong so that she can send files back to the server in Hong Kong when visiting branch offices.

(e) To construct the web page, David uses a text editor to write HTML code instead of using a web authoring tool. Give one benefit of each tool.

Text editor: _____

_____

Web authoring tool: _____

_____

(2 marks)

2015

3. John has a tablet computer with the following specifications:

1 GB main memory, 32 GB flash memory, 10" touch screen and microphone

(a) Other than using a virtual keyboard, suggest a way for John to input data without any extra device.

_____

_____

(1 mark)

(b) The tablet computer is equipped with flash memory instead of a small hard disk. Give **two** advantages of using flash memory over a hard disk.

_____

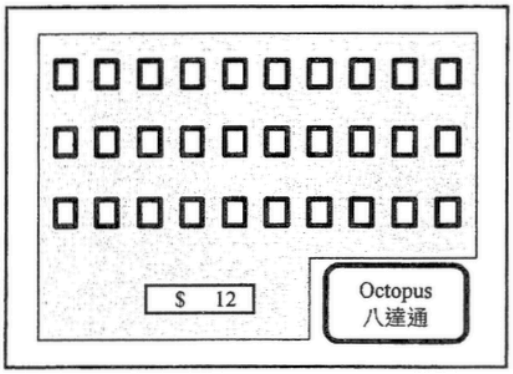
_____

_____

_____

(2 marks)

4. Ms Wong develops a vending machine which has an embedded computer. She writes a program to control the sales of 30 kinds of products in the vending machine.



(a) (i) Other than the operating system, what will the secondary storage in the embedded computer store?

_____

_____

(ii) The operating system is not installed in the ROM of the embedded computer. Why not?

_____

_____

2016

(c) (i) Give **two** settings of the scanner that will affect the quality of the scanned photos.

_____

_____

(ii) Amy realises that she does not need optical character recognition (OCR) software to process the scanned photos. Why not?

_____

_____

(3 marks)

3. Eva works in a trading company. She needs to acquire accounting software for her colleagues to use over the network in the company. She has three solutions:

Solution P: Buy and install standalone accounting software.
Solution Q: Subscribe to the cloud services of an accounting software provider.
Solution R: Develop a new accounting system.

Eva wants to use Solution R.

(b) (i) When purchasing a server, Eva considers the items in the following table. Give a function of each of the second and third items.

Item	Function
Hard disk	Store program data and user data.
Memory cache	
Bus system	

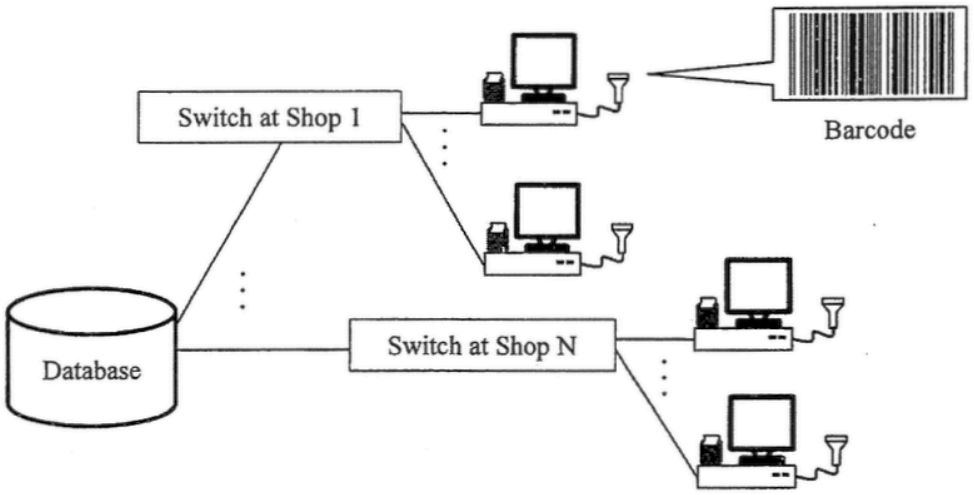
(ii) Give **two** major hardware items that Eva should also consider.

_____

_____

(4 marks)

4. Mary is the IT manager of a toy shop chain. She sets up a point-of-sale (POS) system so that each shop has terminals with barcode readers and printers.



(iii) Which part of the POS system stores the essential information in (a)(ii)?

_____

(iv) Mary chooses thermal printers instead of dot-matrix printers for the POS system. Give **two** reasons to support her choice.

_____

_____

_____

(5 marks)

5. Amy and Billy work in a company. Read their conversation and answer the questions below.

Amy: A new wearable device is very popular in the market. It is like a watch and it can communicate with a smartphone. We can develop a mobile health application for this device.

Billy: I am concerned about the storage capacity of the device and the transmission media. It should support Bluetooth and Wi-Fi.

(a) (i) What kind of memory storage should be used in the device? Why?

_____

_____

(iii) Other than storage capacity and transmission medium, give **two** major design considerations of the wearable device.

_____

_____

2017

(b) The data in the kiosks are stored in Electrically Erasable Programmable Read Only Memory (EEPROM), which is a form of flash memory.

(i) Give one difference between flash memory and RAM.

_____

_____

(ii) Give one difference between flash memory and ROM.

_____

_____

(2 marks)

Again, lets don't waste the blank space <3

(b) The shopping web site sends a promotion email to a group of customers including Eva. Eva receives the email shown below:

From: no-reply@shopshop.com.hk To: cc: Subject: Discount coupon
Dear Customers, Please click the following link to collect the discount coupon: https://shopshop.com.hk/collect.php?ac=1234567890 Customer Services Department

(i) What is the major difference between 'To' and 'cc' (carbon copy) when sending emails to different recipients?

_____

_____

_____

_____

(ii) In this group email, Eva cannot see the email addresses of other recipients. Why not?

_____

_____

2018

A secondary school develops an integrated online student system and provides a number of workstations on the campus for students to use to access the system.

(a) The student system was designed mainly for learning English.

(i) There are two options below for installing the input devices of the workstations. Give an advantage of each option.

Touch screen: _____

_____

Keyboard and mouse: _____

_____

(2 marks)

(ii) Other than display panels and printers, suggest an additional output device for the workstations and justify its use.

_____

_____

(1 mark)

(b) Network interface cards, network cables and switch[illegible]
explain the uses of th[illegible]

(ii) Some users install the program on their mobile devices and play the game for a long time every day, leading to some health problems. Suggest **two** ways to relieve these health problems, other than taking breaks.

_____

_____

_____

_____

(2 marks)

2019

(c) Give **three** suggestions for setting up the game booths so as to reduce the health hazard to participants arising from the use of desktop computers in the booths.

(3 marks)

(c) Peter will execute the program millions of times with `N >= 64` on his desktop computer and wants to improve the performance of the program execution significantly.

(i) Increasing the size of RAM does not help. Why not?

(1 mark)

(ii) Suggest a hardware component that should be upgraded.

(1 mark)

2020

(b) Describe two things John should consider when setting up a new lighting system and curtains in the clinic with reference to ergonomics of the workspace for using computers.

(2 marks)

(c) Give **two** technical reasons why a thermal printer is commonly used instead of a laser printer in self-service kiosks in fast food restaurants. (2 marks)

(e) (i) In which type of memory, RAM or ROM, should the information about the pixels in the display be stored? Give a reason to support your choice.

2021

(b) Mrs Wong uses her desktop computer for more than 6 hours a day.

(i) State the health hazard due to the use of a mouse.

_____

_____

(1 mark)

(ii) Suggest a product for reducing this health hazard.

_____

_____

(1 mark)

(c) John uses his tablet computer 4 hours a day. Give **two** good practices for John to reduce the health hazard of this.

_____

_____

_____

_____

(2 marks)

The school buys two types of tablet computers, P and Q, with the following specifications for students to use:

Specification	P	Q
CPU	5 GHz, 10 cores	1 GHz, 4 cores
RAM	16 GB	4 GB
Storage device	512 GB SSD	16 GB SSD
Screen size	12-inch touch screen	7-inch touch screen
Screen resolution	2736 x 1824	1024 x 600
Connectivity	Bluetooth WiFi	Bluetooth WiFi + Cellular
Other	Physical keyboard 1.2 kg weight Microphone	260g weight Microphone

(b) Describe **two** differences in characteristics between RAM and SSD.

_____

_____

_____

_____

(2 marks)

(c) The CPU specifications of P are 5 GHz and 10 cores. Describe what they represent respectively.

(2 marks)

(d) Without additional devices, suggest **two** ways that users input text in Q.

(2 marks)

(e) Students choose Q instead of P to conduct a survey on the street. Give **two** reasons to support their choice.

(2 marks)

(f) School file servers, P and Q are installed with different operating systems, but files can be transferred between the devices through the network. Why?

(2 marks)

2022

1. Peter decides to buy a computer with a quad-core CPU instead of a dual-core CPU.

(a) (i) Why does Peter make this decision?

_____

_____

(1 mark)

(ii) Suggest another CPU technical specification that he might consider.

_____

(1 mark)

(b) Peter will use his computer to edit and render videos. Give **two** hardware components that he should consider, other than the CPU. Explain briefly.

_____

_____

_____

_____

(2 marks)

(ii) He uses free cloud storage for storing video files. Give an advantage and a disadvantage of using the cloud storage over a portable hard disk.

_____

_____

_____

_____

(2 marks)

2023

1. Mary plans to upgrade her laptop computer.

   (a) Give **two** benefits of replacing the hard disk with an SSD as the secondary storage of the computer.

   _____

   _____

   _____

   _____

   (2 marks)

   (b) Mary plans to increase the size of RAM to enhance the computational power of the computer. What is the main function of RAM? How can it enhance the computational power of the computer?

   _____

   _____

   _____

   _____

   (2 marks)

評卷參考

2012

(a) Notebook computer: hard disk 1
Table computer: flash memory / SSD 1

(b) (i) The CPU / storage device / display unit consumes less electricity. 1

(ii) The operating system (Android) / storage device (flash memory) supports the shorter startup time. / 1
The tablet computer has fewer peripheral devices to load up and thus its startup time is shorter.

(iii) Peter cannot keep his eyes / neck / hands in a good ergonomic position. / small keyboard 1

(c) USB: The printer has a higher data transfer rate / better data security / less interference (stable). 1
Wi-Fi: The printer is more portable. 1

2013

(a) (i) Advantage: It is easy to control / set up the printer. 1

Disadvantage: The printer can be used only when the desktop computer is on. / The desktop computer runs slower when the printer is printing. 1

(ii) (1) data of the document 1
(2) printer's firmware / startup program 1

(c) (i) It has a higher data transfer rate. / It consumes less electricity. / It is smaller in size. / It has a better shock resistance. / It is lighter. / It is more durable. 1×2

2014

(b) It has a higher computational power and a larger screen. 1, 1
(Stable/faster data transfer rate / easier for typing / more programs can be installed)

(ii) Email: Students can submit the reports anytime and anywhere with Internet access. / It is simple because no special network (access right) setting is required. / Email is cross-platform. / Email can provide the identity of the student. / Email can include remarks/descriptions of the reports. 1

School network drive: It is easy for teachers to retrieve all reports at once. / It provides a larger storage capacity / supports larger file size. / The file management is centralised. 1

(e) Text editor: Text editor is available in all OS. / The codes of the web page can be concise. 1

Web authoring tool: Built-in tools/functions facilitate the work. / WYSIWYG / templates provided / user-friendly 1

2015

Marks

3. (a) voice / speech recognition, sound recognition, handwriting, voice input, barcode / QR code scanning (input certain kind of text with a built-in camera and OMR) 1

× siri / voice control / voice-to-text function
× microphone / touch-screen

(b) 1 mark for an answer in each category, maximum 2 categories: 1×2
(1) more portable (lower weight / smaller physical size) (portability)
(2) higher data transfer rate / higher disk access speed / lower latency time / lower seek time
(3) more resistant to shock
(4) quiet in operation
(5) lower power consumption / less heat

× higher running speed
× non-volatile
× more robust (need to be specific)
× higher storage capacity
× lower cost
× The capacity can be increased easily.
× It can be plugged into another computer easily.

4. (a) (i) It stores the program that monitors the sales of the products. / Drivers 1

(ii) It is not easy to update the operating system. 1

2016

(c) (i) Resolution (dpi/ppi), colour depth, file format (compression ratio) 1×2

(ii) There is no character in the photos to be recognised. 1

(b) (i) Memory cache: Store frequently-used program instructions/data. 1
Bus system: It is a communication system that transfer data between major components of the server. 1

(ii) RAM, CPU, RAID controller, NIC 1×2
✖ROM

(iii) Database of the POS system 1

(iv) quiet in operation / smaller in size / shorter printing time 1×2

5. (a) (i) Flash memory should be used as it is light in weight. 1+1
(small in size, anti-shock, portable or low power consumption)

(iii) The weight of the device 1×2
The battery life of the device

2017

(b) (i) Flash memory (EEPROM) is non-volatile. 1

(ii) Flash memory (EEPROM) is re-writable. 1

(b) (i) Senders expect actions and attention from the recipients in 'To' 1
while the recipients in 'Carbon Copy' receive the messages for information only. 1

(ii) The sender has put the email addresses of all recipients in 'bcc'. 1

2018

(a) (i) Touch screen: It is easy for students to operate. / It occupies less space. / It makes for easier hand-eye coordination. 1

Keyboard and mouse: The way that students could input is diversified (e.g. mouse-over) / The input speed could be faster. / The cost is lower. / It is more ergonomic friendly. 1

(ii) Speaker (headphone) – Pronounce English vocabularies. / Play videos. 1

(ii) Control the brightness of the screen. (filter)
Enlarge the font size. (accessibility function)
Use voice input/output. 1×2

2019

(c) Lighting environment when using the computer 1
Posture in using the computer 1
Duration of using the computer 1

(c) (i) The algorithm does not need much memory to be executed. 1

(ii) CPU 1

2020

(b) The lighting system should be set to a suitable brightness to reduce visual discomfort. 1×2
The curtains and/or the lighting system should be set to avoid the monitor to reflect the light.

(c) Fast in printing
No need to replace toner
Smaller in size 1×2

(c) (i) RAM should be used as it is volatile and its content can be over-written.

2021

(b) (i) Repetitive use of wrist / finger / muscle / tendons of arm for data entry 1

(ii) wrist rest / mouse with the ergonomic design 1

(c) Take short breaks periodically during the use. / 1×2
Maintain a good posture. /
Use an anti-glare screen cover. /
Turn on the night shift mode to filter blue light. /
Look at distant objects. /
Use an ergonomic keyboard / mouse.

(simply 'look at green objects' ✗, unless the answer includes distant objects like trees)

(b) RAM: volatile; higher data transfer rate; smaller storage size; CPU is able to access directly; holds program/data currently being executed by the CPU 1×2
SSD: non-volatile; lower data transfer rate; larger storage size; CPU is not able to access directly; holds program/data for long-term storage

(c) 5 GHz: the clock rate of CPU 1
10 cores: 10 processing units within a CPU 1

(d) Virtual keyboard, speech recognition, handwriting recognition 1×2
Alternatives: on-screen keyboard, voice recognition, speech-to-text

(e) allow Internet access via SIM card when WiFi is not available 1
lighter in weight to increase portability 1
✓ cellular network/mobile data (as an alternative to via SIM card)
✓ smaller in size (as an alternative to lighter in weight)
✗ lower power consumption

(f) When the operating systems of all the devices use the same communication protocol for data communication, files can be transferred successfully. 2
✓ FTP (as a specific example of the same communication protocol)
✓ TCP/IP (as a specific example of same communication protocol)

2022

1. (a) (i) It can execute instructions faster (shorter processing time). / 1
It has more computational power. /
It performs more tasks simultaneously (distribute the workload)

(ii) Clock rate / word length (word size) 1

(b) RAM / Hard disk (SSD) / GPU with a simple but reasonable explanation 1×2

(ii) Advantage: 1
File sharing is simpler / Files can be accessed anywhere without bringing the hard disk. /
It can be simultaneously accessed by different devices. / Accessible regardless of OS / It can be available free-of-charge / Easy upgrade

Disadvantage: 1
The access time is longer. / Require Internet access.

2023

1. (a) Faster seek time / shock resistance / lighter in weight / less heat produced during operation / less power consumption / smaller in size 1×2

× more silence in operation (not related to the scenario)

(b) RAM temporarily stores user programs / data 1
Increasing the size of RAM can decrease time needed to access the secondary storage devices. 1